PACLAND · Climate

The verification discipline applied to climate modelling — a plain-English brief

The headlines

‣ The error bar is the size of the signal — 2 models leak heat at ~1.9× the energy imbalance they are used to project.

‣ A 2100 projection is extrapolation, not verification — 0 future states are measured (m = 0).

‣ The leak lives in the atmosphere, not the ocean — 12 of 19 models leak energy, 6 of 19 leak ocean mass.

‣ For the first time, a climate model's trustworthiness is a machine-checked verdict — not a footnote in a paper.

1 · What we are measuring

Climate projections are the most consequential predictions we make. The discipline asks the question the field has never answered in numbers: how much of a projection is verified, versus extrapolated?

2 · The verification deficit

- 4 of 12 historical observables are measured + computable + validated — a certificate radius of 68%.
- 0 of 5 2100-projection targets are validated — the projection is extrapolation, not verification.

A 1% certificate needs ~18,000 validated observables — three orders of magnitude beyond what the historical record can supply. The bottleneck is measured data, not compute.

3 · The energy budget, on real data

- The 0–2000 m ocean alone under-closes the budget (residual 0.15 W/m²).
- The full budget — ocean + atmosphere + land + cryosphere — closes it (residual 0.01 W/m²). An invariant is only an invariant when all its terms are counted.

4 · The CMIP6 conservation drift

- 12 of 19 CMIP6 models leak energy (net TOA drift beyond 0.1 W/m²).
- 2 leak faster than the Earth's energy imbalance: CNRM-ESM2-1 at 1.485 W/m² (1.93× the 0.77 W/m² EEI).
- The leak lives in the atmosphere: median total TOA drift 0.241 W/m² vs 0.051 W/m² in the ocean.
- 6 models also leak ocean mass beyond 0.3 mm/yr — ACCESS-ESM1-5 at -1.03 mm/yr, 30% of observed sea-level rise (3.4 mm/yr).

5 · Why it matters

A model leaking heat at ~2× the signal it is projecting carries an error the size of the projection. Conservation is not a solved detail — it is a hard, quantified defect in most models, and it is invisible to

skill scores and to ensemble spread.

6 · Reproduce

pacland climate · python3 scripts/climate_cmip6_drift.py · docs/climate-cmip6-drift.md